

Adversarial Disturbances for LDS Controller Verification

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Motivation

Design a **provably adversarial** disturbance generator to verify closed-loop controller behavior.

- **Computationally efficient**
- **Task-oriented** certificate of closed-loop controller performance
- Application to **robust training** protocols for learned policies
- Regret guarantee against **rich disturbance policy class**

Generating Adversarial Noise

- (**Stabilizable**) dynamics: $x_{t+1} = Ax_t + Bu_t + Cw_t$
- **Changing quadratic costs** $c_t(x, u) = \|x\|_{Q_t}^2 + \|u\|_{R_t}^2$.
- Parameterize disturbances as $w_t = \sum_{i=1}^H M^{[i]} u_{t-i}$
- Goal: Play disturbances that **minimize regret**

$$\text{Regret}_T(\mathcal{A}) = \max_{\pi \in \Pi} \sum_{t=1}^T c_t(x_t^\pi, u_t) - \sum_{t=1}^T c_t(x_t^A, u_t).$$

Role of **control** and **disturbance** in nonstochastic control are **swapped**, resulting in **non-convex maximization** problem.

Related Work

- Our work builds of the theory of **nonstochastic control** [1] and **online non-convex optimization** [2,3]
- Efficiency follows from **trust region methods** [4]
- **Adversarial Reinforcement Learning (RL)**:
 - In general, there exist many **adversarial agents** in **trained RL** policies, as shown in [5]
 - Heuristic adversarial agents are used in [6] to generate more **robust control policies**, while [7] incorporates heuristic **disturbances into training**

Main Algorithm

Algorithm 2 Memory Online Trust Region (MOTR) Generator

Input: Rounds T , system parameters (A, B) , noise parameter η , history H
for $t = 0$ **to** T **do**

Observe quadratic reward c_t and earn $c_t(x_t, u_t) = \|x_t\|_{Q_t}^2 + \|u_t\|_{R_t}^2$

Generate disturbance $w_t = \sum_{i=1}^H M_t^{[i]} u_{t-i}$

Observe control u_t and update state $x_{t+1} = Ax_t + Bu_t + w_t$

Define $g_t(m) = c_t(y_t(M), u_t)$ where

$$y_{s+1}(M) = \begin{cases} Ay_s(M) + Bu_s + C(\sum_{i=1}^H M^{[i]} u_{s-i} + w_0) & s \geq t - h \\ 0 & \text{otherwise} \end{cases}$$

Define $S_t = S_{t-1} + (\nabla^2 g_t)(0)$ and $s_t = s_{t-1} + (\nabla g_t)(0)$

Generate random vector $\sigma_t \in \mathbb{R}^{Hd_x d_u}$ such that $\sigma_{t,i} \sim \text{Exp}(\eta)$

Update $m_{t+1} \leftarrow \text{TrustRegion}(S_t, s_t - \sigma_t, D, \frac{1}{T})$

Reshape $M_{t+1} \leftarrow \text{reshape}(m_{t+1}, \mathbb{R}^{Hd_x \times d_u})$

end for

Main Theorem

Our algorithm, **MOTR**, achieves $\mathcal{O}(\text{poly}(\mathcal{L})\sqrt{T})$ regret for known, stabilizable dynamics, where $\mathcal{L}$ is a measure of the complexity of the system and comparator class.

Analysis

- Generalize **Non-convex FPL** [2,3] to **Online Learning with Memory** setting. Cost of memory in loss functions can be bounded using **stability** of FPL iterates.
- Introduce the **Online Trust Region with Memory (OTR)**, applying non-convex FPL with memory to quadratic loss functions. Subproblems are solved **efficiently** using **trust region solver** [4].
- Use **fixed horizon rollout** of dynamics to approximate state as linear function of policy matrices M and apply OTR on surrogate costs

Experiments

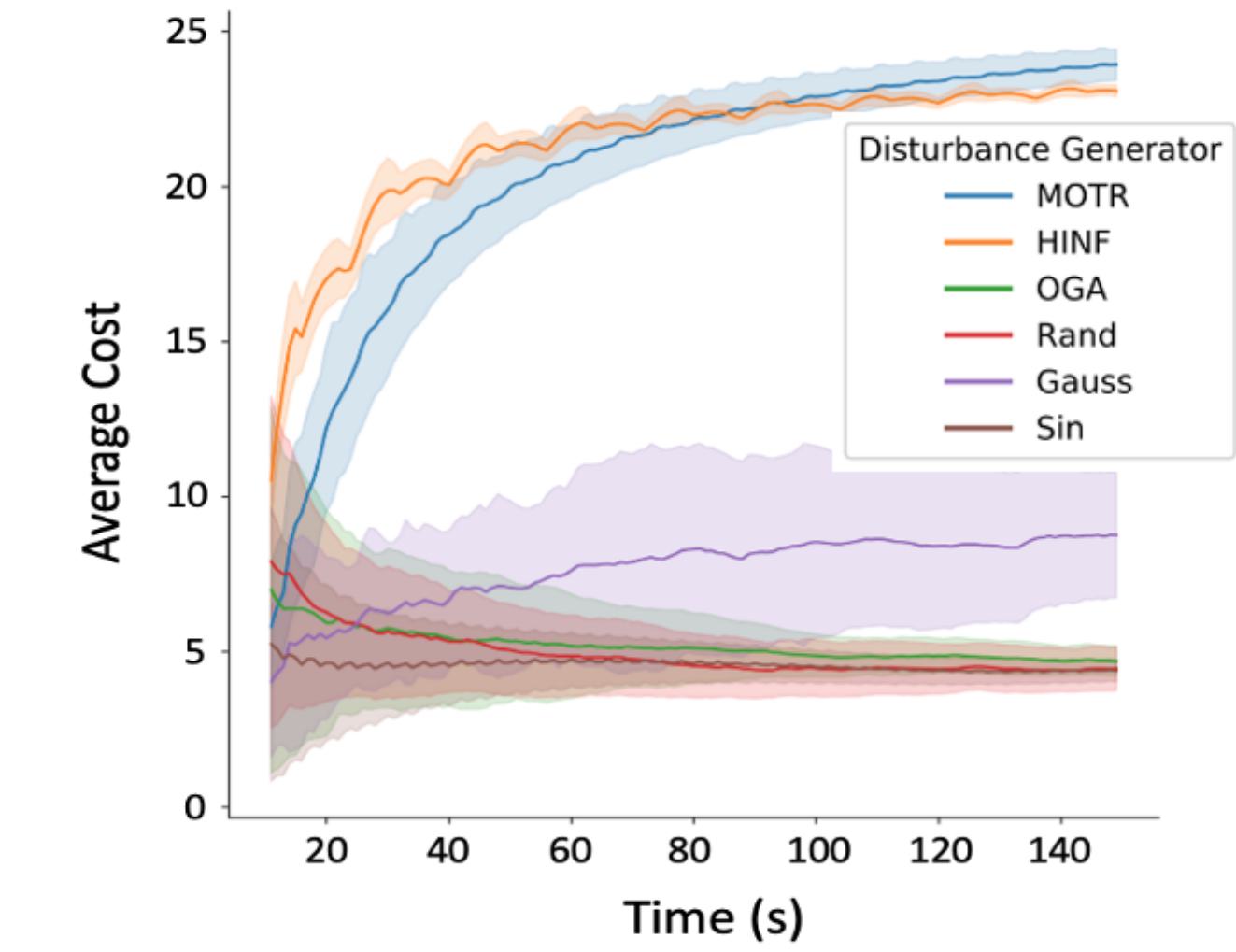
- Validate on **synthetic linear systems** - recover H_∞ Nash equilibrium performance and outperform against other controllers

	LQR	GPC	H_∞
MOTR	1.000	1.000	0.997
OGA	0.918	0.897	0.998
H_∞	0.980	0.949	1.000
Random	0.328	0.323	0.552
Sine	0.564	0.540	0.767
Gaussian	0.444	0.438	0.744

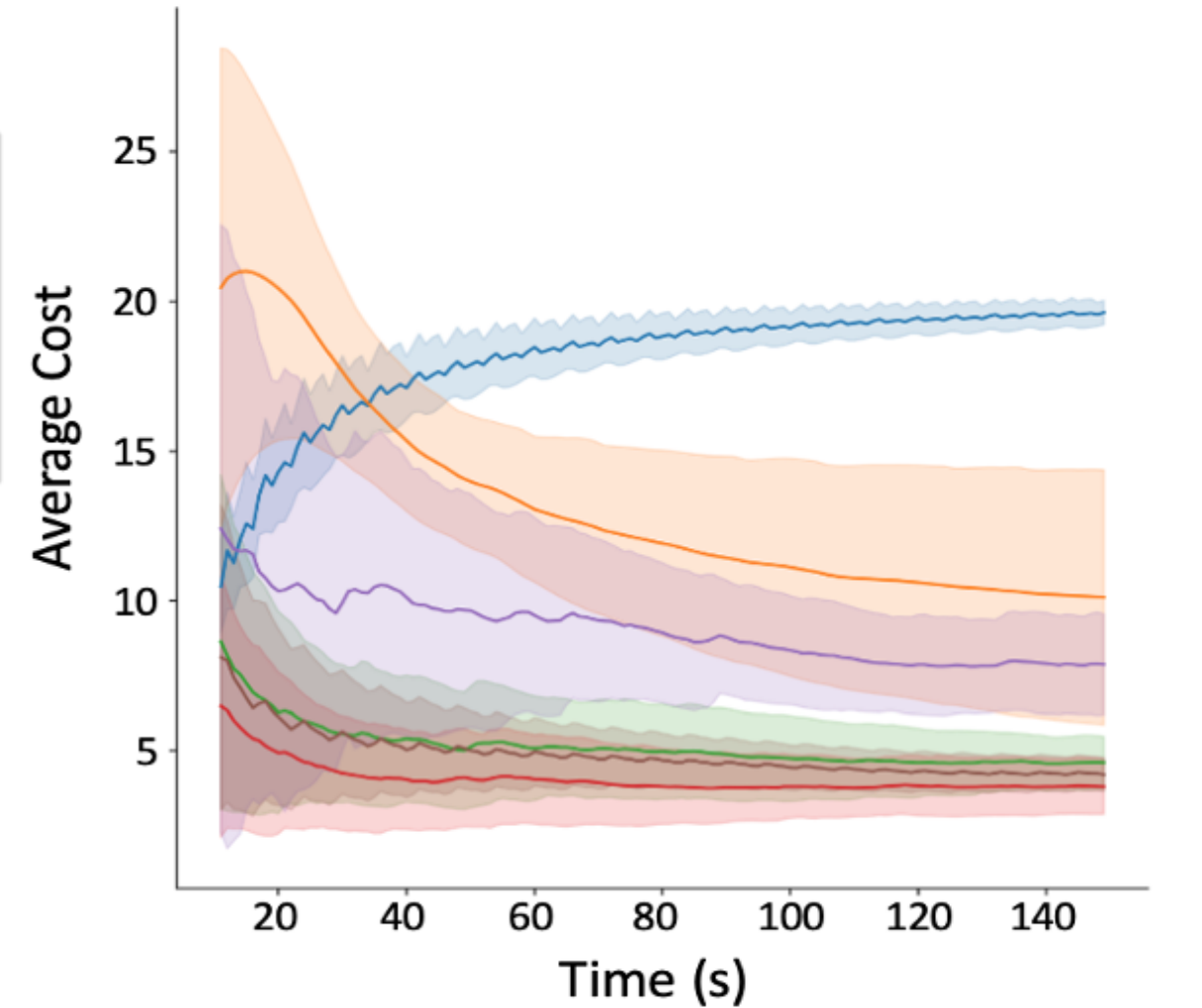
■ Controllers
■ Disturbance Generators

- Test against **SimpleFlight** and **PX4** cascaded PID controllers within **AirSim** environment

SimpleFlight Robust PID Controller



PX4 Cascaded PID Controller



- MOTR recovers **robust behavior** against **SimpleFlight**, and adaptively **outperforms** other baselines against **PX4**
- Outperforms Gaussian and sinusoidal noise.

References (More in Paper)

- [1] Naman Agarwal, Brian Bullins, Elad Hazan, Sham M Kakade, and Karan Singh. Online control with adversarial disturbances. ICML, 2019.
- [2] Naman Agarwal, Alon Gonen, and Elad Hazan. oracle. Learning in non-convex games with an optimization oracle. COLT, 2019.
- [3] Arun Sai Suggala and Praneeth Netrapalli. Online non-convex learning: Following the perturbed leader is optimal. ALT, 2020.
- [4] Aharon Ben-Tal and Marc Teboulle. Hidden convexity in some nonconvex quadratically con-strained quadratic programming. Mathematical Programming, 1996.
- [5] Adam Gleave, Michael Dennis, Cody Wild, Neel Kant, Sergey Levine, and Stuart Russell. Adversarial policies: attacking deep reinforcement learning. ICLR, 2020.
- [6] Eugene Vinitsky, Yuqing Du, Kanaad Parvate, Kathy Jang, Pieter Abbeel and Alexandre Bayen. Robust reinforcement learning using adversarial populations. Preprint, under review.
- [7] Ajay Mandlekar, Yuke Zhu, Animesh Garg, Li Fei-Fei, and Silvio Savarese. Adversarially Robust Policy Learning: Active Construction of Physically-Plausible Perturbations IROS, 2017.